Report

The steps that were taken to use the data are as follows:

- First I created individual data frames of all the years, making sure that Rstudio is able to read the CSV files appropriately.

- Next I combined all the individual data frames of each year's data, into a single POARTOTAL dataframe. As well as renaming the column PM.sub.10..sub..particulate.matter..Hourly.measured into “PM10” in the datasets.

- Next in order to display the specified dates data, I had to filter the dates from POARTOTAL. I created individual data frames for each specified dates. As well as ensuring that PM10 doesn’t have any NA values in its column.

- Next I combined the filtered dates again into a data frame TOTAL_DATE.

- I was able to create visualisations 1 - 3. As all the steps were reusable for those visualisations. For visualisation 1 I chose a Plotly scatter graph for all the specific dates as it can show all the trends and is easy to compare dates. For visualisation 2 I chose GGPLOT facet_warp as using it can make it easy to identify the trends of the pollutant and make comparisons. For visualisation 3 I chose box plot as it can show the median, maximum, minimum etc. Which is useful for understanding general data trends.

- For visualisation 4 it is required to display monthly averages in 2020 of all the pollutants. In order to do this. I created a data frame POAR2020_filtered which adds the contents of POARTOTAL dataframe to it and filters for the year 2020 and all its monthly datas. After which, I created another data frame monthly_avg_2020 which adds the contents from POAR2020_filtered dataframe and calculates the averages of all the pollutants while leaving any NA values. Now I can use the monthly_avg_2020 dataframe to create visualisation 4. I chose a heatmap as it can display the average and show which months have high to low concentrations of each pollutant, it is also easy to compare each month's average.

- Visualisation 5 I had to create yearly averages of all the pollutants. In order to do this, I created a dataframe POARTOTALYEARS which takes the POARTOTAL data, and filters it so data for every year is present. Next I created the averages of all pollutants in yearly_avg dataframe. Then I used yearly_avg dataframe to create Visualisation 5. Using a Bar chart is the most effective in showing averages for a lot of data, as well as easy to compare between each pollutant level for each year.

Short Description - Discoveries

Visualisation 1

Visual 1 displays the pollutant known as “PM10”. I found that for the years 2018, 2020, 2022 the PM10 levels are consistently between 10 - 25 PM10 concentrations, throughout the 24 hours of the specified dates. This tells me that those PM10 levels are normal for the area. The dates that stood out to me are 29-11-2021, 03-01-2019 and 26-03-2020. For 26-03-2020 the PM10 concentrations peak the highest of all dates at nearly 50 PM10 concentration, along with throughout the 24 hours of that date, the PM10 levels being overall high. Also for the dates 29-11-2021 and 03-01-2019 there are gaps in data, however it could be that the PM10 levels actually decreased completely and increased back up again, making it seem like there are missing data.

Visualisation 2

Visualisation 2 displays the pollutant “Nitric oxide”. This pollutant follows a similar trend with the PM10 pollutant. Usually varying between 10 - 25 concentrations of Nitric oxide. 29-11-2021 data also makes it seem like there is missing data for Nitric oxide levels. However I think it’s important to consider these blank data points, for other possible reasons as to why the pollutant levels fluctuate the way they do.

Visualisation 3

Visualisation 3 displays another pollutant known as “Nitrogen oxides as nitrogen dioxide”. I found that there are even higher levels of this pollutant present compared to the previous pollutants. 29-11-2021 hitting a maximum of 173 levels of Nitrogen oxides as nitrogen dioxide. The other dates also generally have higher concentrations of this pollutant.

Visualisation 4

Visualisation 4 displays the monthly averages in 2020 of all the pollutants. These pollutants are “PM10”, “Nitric oxide”, “Nitrogen oxides as nitrogen dioxide” and “Nitrogen dioxide”. The heatmap shows Nitrogen oxides as nitrogen dioxide having the highest concentration of all pollutants during January, on average 59. As well as throughout the year, generally having the highest concentration of all pollutants. All the other pollutants have similar varying monthly concentrations. With the lowest concentration being 2.97 average concentration of Nitric oxide during the month of April.

Visualisation 5

Visualisation 5 I found from all the visualisations that the pollutant “Nitrogen oxides as nitrogen dioxide” seems to have higher concentrations compared to PM10, Nitrogen dioxide and Nitric oxide. I wanted to discover whether this trend followed over a longer period of time. Which is why for this visualisation I created yearly averages of all pollutants, in all the years I have the data for. Ranging from 2018 - 2023. Giving me 5 years of data to conclude which pollutant is more present as well as get a general idea of all pollutant levels. Nitrogen oxides as nitrogen dioxide has the highest average concentration of all, followed by Nitrogen dioxide, PM10 and lastly Nitric oxides. This is an important discovery, as it can lead to better understanding of possibly which pollutant can cause more harm then the other, or conversely which pollutant is less harmful then the other to the environment.

References for visualisations

[1] TheGoat. (2016, August 16). Skip specific rows using read.csv in R [duplicate]. https://stackoverflow.com/questions/39110755/skip-specific-rows-using-read-csv-in-r

[2] Akrun. (2021, May 14). Using r: bind rows of multiple dataframes, which may or may not exist. https://stackoverflow.com/questions/67541348/using-r-bind-rows-of-multiple-dataframes-which-may-or-may-not-exist

[3] OpenAI. (2023, October 23). how to use filters in Rstudio and how can i also exclude any NA values. (Sep 25, 2023 version) [Large language model]. https://chatgpt.com/share/67e92e83-c120-8002-94d9-a75b9a88e383

[4] Dream. (2019, October 18). What does na.rm=TRUE actually mean? https://stackoverflow.com/questions/58443566/what-does-na-rm-true-actually-means

[5] Plotly. (2025, March 23). Line Plots in R. https://plotly.com/r/line-charts/

[6] GGPLOT2. (2025, March 25). Wrap a 1d ribbon of panels into 2d. https://ggplot2.tidyverse.org/reference/facet_wrap.html

[7] Plotly. (2025, March 25). Box Plots in R. https://plotly.com/r/box-plots/

[8] OpenAI. (2023, October 23). How to mutate a format dmy into something i want in Rstudio. https://chatgpt.com/share/67e931e9-f44c-8002-a62f-332c03cd54ba

[9] Clayton Samples. (2015, October 15th). How to group by two columns in R https://stackoverflow.com/questions/33152620/how-to-group-by-two-columns-in-r

[10] Dplyr. (2025, March 26). Summarise each group down to one row. https://dplyr.tidyverse.org/reference/summarise.html

[11] Tidyr. (2025, March 26). Pivot data from wide to long. https://tidyr.tidyverse.org/reference/pivot_longer.html

[12] Plotly. (2025, March 27). Heatmaps in R. https://plotly.com/r/heatmaps/

[13] Plotly. (2025, March 27). Bar Charts in R. https://plotly.com/r/bar-charts/